

WEB TECHNOLOGIES

CSE-4004

LECTURE 5: CSS Selectors

By: Dr.MD.Sirajuddin

CSS SELECTORS

These are the patterns used to select HTML elements for applying styles.

1. Type Selector or Tag Selector: Select HTML Tag directly.

Example:

```
p  
{  
color:red;  
}
```

2. ID Selector: Used to select an element with unique ID.

Example:

```
#elementid { color:red; }
```

CSS SELECTORS

3. Class Selector: Selects element(s) that have same class.

Example:

```
.classname { color:red; }
```

4. Universal Selector: Selects all elements.

Example:

```
* { color:red; }
```

5. Child Selector: Selects Children of an element

```
ul > li { color:red; }
```

6. Group Selector: Selects multiple elements

```
h1,h2,h3 { color:red; }
```

selector.css

```
p
{ color:red; }

#elementid { color:red; }

.className { color:blue; }

ul > li { color:yellow; }

h1,h2,h3 { color:green; }
```

```
<html>
<head>
<title> CSS-SELECTOR </title>
<link rel="stylesheet" href="selector.css">
</head>
<body>
<p> Hello </p>
<i id="elementid"> VIT-AP </i>
<ul> <li> Web Technologies </li> </ul>
<h1>SCOPE</h1>
<h2>SELECTOR CONCEPT </h2>
<ol class="classname">
<li>SELECTOR CONCEPT </li>
</ol>
</body> </html>
```

Hello

VIT-AP

- Web Technologies

SCOPE

SELECTOR CONCEPT

1. SELECTOR CONCEPT

<div>: division

- **Used to create a section or division in a webpage.**
- **It is a container used to group elements in a webpage.**

Example

<body>

<div style="background-color:lightblue; width:300px; border-width:8px; border-style:solid; padding:20px;" >

<h2>Welcome Students</h2>

<p>Web Technologies is inside a div box.</p>

</div>

</body>

Welcome Students

Web Technologies is inside a div box.

CSS- COLORS AND BACKGROUNDS

Property	Description	Example
color	Text color	color: red;
background-color	Background color	background-color: yellow;
background-image	Set image as background	background-image: url("pic.jpg");
background-repeat	repeat/no-repeat image	background-repeat: no-repeat;
background-size	Size of background	background-size: cover;
background-position	Position of background	background-position: center;
background-attachment	Fixed, scroll	background-attachment: fixed;

CSS- BOX MODEL

Property	Description	Example
Content	Actual text/image	—
Padding	Space inside border	padding: 10px;
Border	Line around the box	border: 2px solid black;
Margin	Space outside border	margin: 20px;

CSS- MARGINS, PADDING & BORDERS

Property	Meaning	Example
margin	Outside spacing	margin: 15px;
margin-top	Top margin	margin-top: 10px;
padding	Inside spacing	padding: 20px;
padding-left	Left padding	padding-left: 5px;
border	Border size, style, color	border: 3px solid red; border-width:5px
border-bottom	Bottom border only	border-bottom: 2px dashed blue;
border-top	Upper border only	border-top:2px solid red;

CSS- TEXT PROPERTIES

Property	Description	Example
color	Text color	color: green;
text-align	Center/left/right	text-align: center;
text-decoration	Underline/none	text-decoration: underline;
text-transform	uppercase/lowercase/ capitalize/none	text-transform: uppercase;
letter-spacing	Space between letters	letter-spacing: 2px;
line-height	Space between lines	line-height: 1.5;

CSS- FONT PROPERTIES

Property	Description	Example
font-size	Size of text	font-size: 20px;
font-family	Type of font	font-family: Arial;
font-weight	Boldness	font-weight: bold;
font-style	Italic/normal	font-style: italic;
font (shorthand)	All font settings in one line	font: bold 18px Arial;

POSITIONING: Used for positioning an element

Position Value	Meaning	Moves with top/left/right/bottom?	Scroll Behaviour	Description
static (default)	Normal document flow	✗ No	✓ Moves with page	Default position; cannot be moved using top/left.
relative	Relative to its original position	✓ Yes	✓ Moves with page	Slight adjustments without removing from flow. Useful for small shifts.
absolute	Relative to nearest positioned ancestor	✓ Yes	✓ Moves with page	Taken out of normal flow. Used for overlays, icons inside boxes, etc.
fixed	Relative to the viewport (screen)	✓ Yes	✗ Stays fixed	Stays in same location even when the page scrolls. Used for sticky headers, menus.

POSITIONING

```
<style>
```

```
.box {
```

```
  width: 100px;
```

```
  height: 100px;
```

```
  background: orange;
```

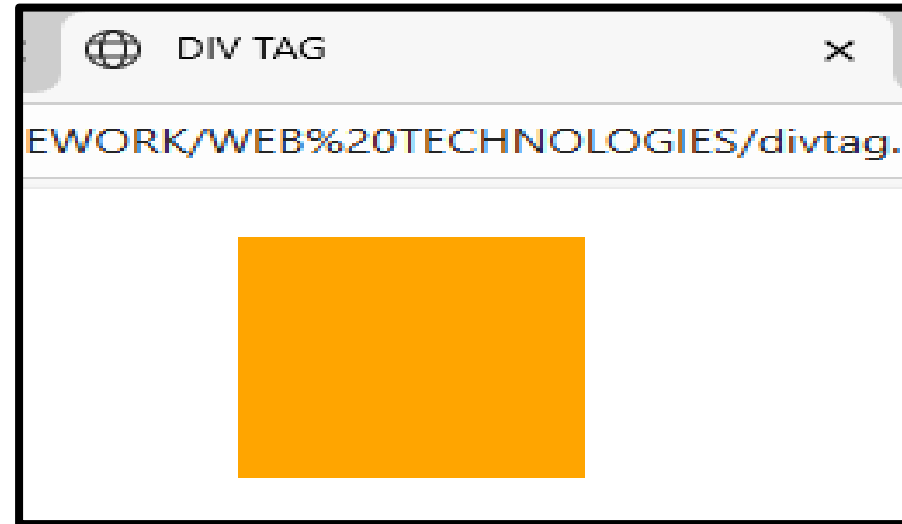
```
  position: absolute;
```

```
  top: 50px;
```

```
  left: 50px;
```

```
}
```

```
</style>
```



Z-index: used to control which element appears on top when elements overlap. Element with higher z-index placed above. It works when element has position value set.

GRID LAYOUT

```
<html>
<head>
<title> Grid </title>
<style>
.chessboard {
  display: grid;
  grid-template-columns: repeat(2, 80px);
  grid-template-rows: repeat(2, 80px);
  grid-auto-flow: row; /* default */
}
.square {
  width: 80px;
  height: 80px;
  border: 1px solid black; }
```

```
.black
{
  background: black;
}

.white
{
  background: white;
}
</style>
</head>
```

GRID LAYOUT

```
<body>  
<div class="chessboard">  
  <div class="square black"></div>  
  <div class="square white"></div>  
  <div class="square white"></div>  
  <div class="square black"></div>  
</div>  
</body>  
</html>
```

